01.

TECHNOLOGY MANAGEMENT | ASSIGNMENT 1

BEYOND MANUAL TESTING

AN AI-DRIVEN GAME TESTING STRATEGY FOR UBISOFT

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UBISOFT

02.

COMPANY OVERVIEW

Ubisoft at a Glance

40	~16.6K	129M	2
Years	Employees	Players	Engines

COMPANY PROFILE

- Industry: Video game development and publishing
- Founded: 1986, France
- Franchises: Assassin's Creed, Far Cry, Rainbow Six

TECHNOLOGY FOUNDATION

- ANVIL Game Engine
- SNOWDROP Game Engine
- LA FORGE Research & AI

03.

INNOVATION HUB

Ubisoft La Forge

— Existing AI Research & QC Applications

ABOUT LA FORGE

Ubisoft's R&D group connects academic research with practical game development through AI, machine learning and technical prototyping.

WHY IT MATTERS

Ubisoft has already explored AI navigation agents and ML-powered bots for Quality Control. Our proposal focuses on expanding these capabilities into a scalable testing strategy.

RESEARCH EVIDENCE / GDC 2024

Building ML Bots at Ubisoft La Forge

From Research to Production
Gabriel Robert · GDC Vault

01 — REINFORCEMENT LEARNING

AI agents learn to navigate complex 3D game environments.

02 — ML-POWERED GAME BOTS

ML bots explore gameplay scenarios and simulate player-like behavior.

03 — QUALITY CONTROL APPLICATIONS

ML bots can support quality control in complex game environments.

04.

PROBLEM STATEMENT

Scaling Game Quality Assurance

01 — COMPLEX GAME WORLDS

Large environments and interconnected gameplay systems create extensive testing possibilities.

02 — REGRESSION TESTING

Game updates require repeated checks to ensure existing features continue to function correctly.

03 — LATE BUG DETECTION

Defects discovered late can increase rework and put development schedules at risk.



CORE TECHNOLOGY CHALLENGE

How can Ubisoft scale game testing without proportionally increasing manual QA effort?

05.

REAL-WORLD CASE STUDY

Assassin's Creed Unity: A QA Lesson



Historical case illustrating the importance of game quality assurance.

01 / WHAT HAPPENED?
The 2014 launch faced bugs, stability issues and performance problems.

02 / UBISOFT'S RESPONSE
Ubisoft publicly acknowledged these problems after release.

03 / STRATEGIC LESSON
Earlier defect detection and stronger regression testing can help reduce release risk.

06.

SOLUTION OVERVIEW / PROPOSED TECHNOLOGY

Proposed AI-Driven Testing Stack

Three complementary capabilities to expand test coverage, automate checks and prioritize QA efforts.

01 / EXPLORE

RL-BASED TESTING AGENTS

Autonomously navigate environments and execute gameplay scenarios.

Focus: Test Coverage

02 / VERIFY

AUTOMATED REGRESSION TESTING

Re-run repeatable tests to check whether updates break existing functionality.

Focus: Reliability

03 / PRIORITIZE

ML-BASED DEFECT PREDICTION

Use historical defect and change data to flag potentially high-risk areas.

Focus: Risk Prioritization

CROSS-CUTTING CAPABILITY

Human QA + Test Telemetry

Automated checks, logs and human review help validate potential defects.

07.

SYSTEM ARCHITECTURE

How AI-Driven Testing Works

01 — BUILD
CI/CD Pipeline

→

02 — PREPARE
Test Orchestration

→

03 — EXECUTE
Game Engine + AI Agents

→

04 — VALIDATE
Telemetry + Automated Checks

→

05 — REVIEW
Defect Triage + Human QA

SUPPORTING CAPABILITIES
Game Engine Adapter · Test Repository · ML Risk Prioritization



VIDEO DEMONSTRATION
AI-Controlled Gameplay Exploration
Conceptual architecture proposed for Ubisoft. Engine compatibility requires pilot validation.

08.

OPTIONS OVERVIEW

Technology Acquisition Alternatives

Five strategic routes to build or obtain Ubisoft's proposed AI-driven testing capabilities.

MAKE

Develop AI testing capabilities internally through Ubisoft La Forge and production teams.

Advantage: Greater control over technology and IP.

Risk: High development effort and longer time to scale.

BUY

Purchase and deploy a commercial testing solution for internal use.

Advantage: Faster access to established tools.

Risk: Customization and engine-integration challenges.

LICENSE

Access third-party testing technology under a defined license or subscription.

Advantage: Flexible access with lower initial commitment.

Risk: Ongoing fees, usage restrictions and vendor dependence.

PARTNER

Co-develop testing capabilities with AI specialists or research partners.

Advantage: Combines external expertise with Ubisoft's R&D.

Risk: Coordination, IP allocation and knowledge-transfer challenges.

ACQUIRE

Acquire a specialized AI testing company to obtain its capabilities and team.

Advantage: Potential access to technology, talent and IP.

Risk: High investment and complex organizational integration.

KEY QUESTION

Which option offers Ubisoft the best balance of technical fit, speed, cost and long-term control?

09.

COMPARATIVE VIEW

Strategic Trade-Off Analysis

STRATEGIC VALUE FOR UBISOFT

The preferred strategy should strengthen Ubisoft's internal AI capabilities while balancing technical feasibility, cost, speed and risk.

STRATEGIC PRIORITY

Build reusable AI testing capabilities across suitable Ubisoft development workflows.

Qualitative assessment by the project team; not measured Ubisoft performance data.

Speed

Buy and License may accelerate piloting if technical integration is feasible. Make generally requires more development time.

Cost

Make requires sustained internal investment. Licensing may reduce initial commitments but introduces recurring costs.

Technical Capability

External solutions must be evaluated for compatibility with Ubisoft's proprietary game engines and QA workflows.

Long-Term Control

Make supports internal technology and IP control. Partnerships offer complementary expertise but require clear agreements.

10.

MARKET SCAN

External Technology Landscape & Internal R&D

modl.ai

AI-POWERED GAME TESTING

Capabilities

- Autonomous gameplay testing
- Visual bug reporting
- SDK-free, black-box testing

Strategic Value

Potential for automated exploratory QA.

Key Unknown

Suitability for complex AAA gameplay, data security and platform support.

GameDriver

Automated test framework

Capabilities

- Unity, Unreal and Godot support
- Repeatable functional tests
- CI/CD automation

Strategic Value

Potential for scalable regression testing.

Key Unknown

Integration depth with Ubisoft's Anvil and Snowdrop engines.

Internal R&D

Ubisoft La Forge prototypes

Capabilities

- AI and machine learning research
- Game bot prototyping
- Research-to-production collaboration

Strategic Value

Strengthens internal technical expertise.

Key Unknown

Resources and effort required to scale prototypes across projects.

EVALUATION NOTE

External vendors are potential candidates, not confirmed Ubisoft partners. Technical compatibility, security and commercial terms require validation.

Sources: modl.ai (2026); GameDriver Technology & Documentation (2026); Ubisoft La Forge.

11.

EVALUATION FRAMEWORK

Technology Selection Criteria

01

COST | 15%

Total cost of development, licensing, integration and maintenance.

02

SPEED | 15%

Time required to pilot, deploy and scale the solution.

03

TECHNICAL CAPABILITY | 30%

Compatibility with Ubisoft's game engines, QA workflows and scalability requirements.

04

STRATEGIC VALUE | 25%

Contribution to Ubisoft La Forge, organizational learning and knowledge transfer.

05

LONG-TERM CONTROL | 15%

IP rights, customization flexibility and vendor dependence.

Weights reflect the project team's proposed priorities for Ubisoft.

12.

SCENARIO ANALYSIS

Weighted Acquisition Decision Matrix

OPTION	COST w 15%	SPEED w 15%	TECHNICAL w 30%	STRATEGIC w 25%	CONTROL w 15%	SCORE
Make	2	2	4	5	5	3.80
Buy	3	5	2	2	2	2.60
License	4	5	2	2	2	2.75
Partner	3	4	4	5	4	4.10
Acquire	1	2	3	4	5	3.10

PREFERRED OPTION: PARTNER / JV — 4.10 / 5

A partnership-led approach achieves the highest weighted score, combining strong technical capability with strategic value for Ubisoft.

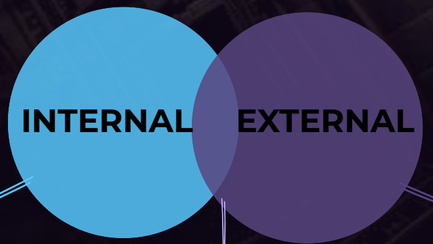
Illustrative team assessment · Ratings: 1–5 (higher is better) · Subject to technical validation.

13.

RECOMMENDED ACQUISITION STRATEGY

Hybrid Strategy — Internal R&D + External Partnership

Combining Ubisoft's internal AI expertise with external testing capabilities.



UBISOFT LA FORGE + QA

- AI research and testing agents
- Engine integration and game expertise
- Internal capability development

JOINT PILOT

Validate AI-assisted testing on selected game scenarios.

EXTERNAL TECHNOLOGY PARTNER

- Complementary testing tools
- Automation and integration support
- Joint technical training & documentation

DECISION GATE — VALIDATE BEFORE SCALING

Proceed only if engine compatibility, QA effectiveness and IP safeguards are confirmed.

14.

INNOVATION ENGINE

Corporate R&D Strategy — Ubisoft La Forge

Build on La Forge's existing ML-bot research to develop and validate AI-assisted game testing capabilities.

AI RESEARCH & ADAPTATION

Adapt machine-learning bot research to execute targeted gameplay testing scenarios.

ENGINE-LEVEL PROTOTYPING

Prototype agent interfaces, telemetry and automated checks with Ubisoft engineering teams.

JOINT EXPERIMENTATION

Collaborate with QA teams and potential technology partners to evaluate test coverage and reliability.

R&D GOVERNANCE & VALIDATION

Assess technical results, integration effort and IP safeguards before deciding whether to scale.

R&D OBJECTIVE

Deliver a validated AI-assisted QA prototype and an evidence-based scale-up decision.

Proposed R&D responsibilities, not a confirmed Ubisoft implementation.

Sources: Ubisoft La Forge; GDC 2024 — Building ML Bots at Ubisoft La Forge.

15.

ABSORPTIVE CAPACITY

Technology Transfer & Capability Building

Enable La Forge and Ubisoft QA teams to adopt external testing technology through documentation, joint learning and hands-on training.

01
DOCUMENT & SECURE

Agree usage and IP terms; document tool interfaces, test assets and workflows.

02
JOINT PILOT & SHADOWING

Partner, La Forge and QA engineers configure, test and troubleshoot together.

03
TRAIN & HAND OVER

Train QA leads; transfer test suites, runbooks and support procedures.

04
EMBED & IMPROVE

Assign studio owners; use feedback to refine tools and training.

EXPLICIT KNOWLEDGE

Technical documentation · Reusable test cases

TACIT KNOWLEDGE

Workshops · Shadowing · Paired engineering

ABSORPTIVE CAPACITY

QA champions · Hands-on practice · Feedback

Proposed technology transfer mechanism; no existing vendor agreement is implied.

16.

RISK & COMPLIANCE

Intellectual Property, Data & AI Governance

Protecting proprietary technology and maintaining oversight of the proposed AI testing partnership.

IP & LICENSING RIGHTS

Protect Ubisoft-owned game assets and define rights to existing AI models and co-developed testing tools.

Control: Written IP and licensing agreements.

DATA SECURITY & PRIVACY

Restrict access to pre-release game builds, source code and testing data.

Control: Access restrictions and secure testing environments.

PARTNER & CONTRACT GOVERNANCE

Define permitted technology use, confidentiality, support obligations and exit rights.

Control: Vendor due diligence and contractual safeguards.

AI TESTING GOVERNANCE

Monitor test reliability, false positives and traceability of AI-generated findings.

Control: Audit logs and human QA verification.

Proposed governance measures, subject to legal and technical review.

17.

PHASED ROLLOUT

Proposed 12-Month Implementation Roadmap

Four phases to evaluate, integrate and selectively scale Ubisoft's proposed AI-assisted testing capability.

MONTHS 1–3

SCOPE & SELECT

Define pilot scenarios, baseline QA metrics and select a potential technology partner.

MONTHS 4–6

CO-DEVELOP & PILOT

Assess engine compatibility, test accuracy, security and CI/CD integration.

Months 7–9

VALIDATE & INTEGRATE

La Forge, QA teams and the partner test AI agents on a selected game build.

Months 10–12

SELECTIVE ROLLOUT

Expand to approved QA workflows, train teams and monitor performance.

GO / NO-GO — AFTER VALIDATION

Proceed to rollout only when technical compatibility, QA effectiveness and IP requirements are satisfied.

18.

TARGET OUTCOMES

Expected Benefits & Performance KPIs

Measuring the potential value of AI-assisted game testing.

01

WIDER TEST COVERAGE

KPI: Automated Scenario Coverage

Share of priority gameplay scenarios executed automatically.

02

LESS REPETITIVE QA WORK

KPI: Manual Regression Hours

Hours spent on repeatable manual tests per game build.

03

FASTER TESTING FEEDBACK

KPI: TIME TO QA RESULTS

Time from a new game build to actionable test results.

04

RELIABLE AI FINDINGS

KPI: QA-CONFIRMED FINDING RATE

Share of AI-reported issues confirmed by human QA.

PILOT MEASUREMENT APPROACH

Establish baseline metrics before the pilot; compare results and set scale-up thresholds after evaluation.

19.

THE DECISION

Final Strategic Recommendation

Build on La Forge.
Partner selectively.
Scale only after validation.

DECISION BASIS

Partner/JV achieved the highest weighted score: 4.10/5 in our illustrative evaluation.

NEXT ACTION

Select a suitable technology partner, launch a limited pilot and validate engine compatibility, QA effectiveness and IP safeguards before scaling.

20.

SOURCES

References

UBISOFT & INDUSTRY EVIDENCE

[1] Ubisoft. (n.d.). Ubisoft La Forge. Official R&D overview.
[2] Ubisoft. (n.d.). Technology & Innovation. Official game-engine and technology overview.
[3] Robert, G. (2024). AI Summit: Building ML Bots at Ubisoft La Forge: From Research to Production. Game Developers Conference.
[4] Ubisoft. (2014). Performance Optimizations Currently Being Tested — Assassin's Creed Unity. Official Steam community announcement.

TECHNOLOGY PROVIDERS & ACADEMIC RESEARCH

[5] modl.ai. (n.d.). AI-Powered Game Testing. Official product website.
[6] GameDriver. (n.d.). GameDriver Knowledge Base. Official technical documentation.
[7] Gillberg, J., et al. (2023). Technical Challenges of Deploying Reinforcement Learning Agents for Game Testing in AAA Games. IEEE Conference on Games.
[8] Cohen, W. M., & Levinthal, D. A. (1990). Absorptive Capacity: A New Perspective on Learning and Innovation. Administrative Science Quarterly, 35(1), 128–152.

LET'S DISCUSS

Thank you ~~
Questions?

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